

PROPOSAL FOR DISCUSSION

TrueYield Jharkhand

Revenue leakage analytics across departments: one view of every taxpayer, and a verified list of revenue the state is already owed

Prepared for: **Finance Department, Government of Jharkhand**

With: Commercial Taxes · Mines & Geology · Revenue, Registration & Land Reforms · Transport · Excise & Prohibition · Energy (JBVNL)

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4-month pilot

3 departments · 3 districts · read-only

₹56.5 lakh

pilot, one-time

Two engines

detection (built) + ML discovery

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Executive summary

Jharkhand expects ₹45,999 crore of own tax revenue in 2026–27 and about ₹16,000 crore from mining. Every one of these rupees is collected by a department that sees only its own records. The Commercial Taxes Department sees GST returns but not the electricity a business draws; Mines & Geology sees royalty but not the stamp duty paid on the lease deed; Registration sees a sale deed but not the turnover of the buyer. Leakage lives in the gaps between these registers, and CAG audits of Jharkhand have repeatedly found it there.

TrueYield is a revenue-leakage analytics platform. It brings each department's records into a common data lake on a regular schedule, compares them for the same taxpayer, and produces a ranked, rupee-quantified list of cases for officers to verify. A detection engine applies known rules; a discovery engine uses machine learning to find new patterns and turn them into rules. It does not replace any departmental system and does not assess anyone; it tells officers where to look and shows the arithmetic behind every figure.

THE PROPOSAL IN FIVE LINES

- **What:** a cross-department leakage platform for the Finance Department, starting with Commercial Taxes, Mines & Geology and Registration.
- **Why now:** our review of public information found no cross-department revenue-leakage platform in Jharkhand today; GST-only national analytics give the state limited access and do not cover mining, stamps, land, transport or excise.
- **Proof:** the detection engine is our product, already running on test data with a library of 85 checks across 10 families; it is licensed and adapted to Jharkhand, while the state-scale data platform and the ML discovery engine are built for Jharkhand.
- **Pilot:** 4 months, 3 departments, 3 districts, read-only access, ₹56.5 lakh. Full rollout only if the pilot's verified results justify it.
- **Total:** ₹1.97 cr one-time for the full state rollout including the platform licence, data platform and discovery engine, and ₹47.5 lakh a year to run.

What the Government gets

- A single view of each taxpayer across GST, mining, stamps and registration, land, vehicles, excise and electricity.
- A ranked verification list for each district and department, every case with its evidence, arithmetic and statutory rate.
- Clear separation between evidence (both records are the state's own, keyed to the same taxpayer) and leads (patterns that need field verification).
- Recovery tracking from assignment to realisation, and a leadership dashboard for the Finance Secretary.
- New rules found by machine learning, reviewed by officers before they are used.
- Departments' own systems are untouched; data is copied read-only into a secure data lake in the Jharkhand State Data Centre.

1 JHARKHAND'S REVENUE AND WHERE IT LEAKS

Jharkhand's own revenue rests on a few large heads, each administered by a different department with its own register.

Revenue head (2026–27 BE)	Estimate	Administered by / main register
Own tax revenue (total)	₹45,999 crore	Finance Department (all heads below)
State GST	₹14,563.75 crore	Commercial Taxes: GST returns, e-way bills
Taxes on sales, trade etc. (VAT on petroleum, liquor)	₹7,899.72 crore	Commercial Taxes
State excise	₹4,500 crore	Excise & Prohibition; JSBCL (wholesale)
Stamps & registration	≈ ₹1,800 crore	Revenue, Registration & Land Reforms: NGDRS / Jharnibandhan
Mining receipts (non-tax)	≈ ₹16,000 crore	Mines & Geology: JIMMS

Sources: PRS Legislative Research, Jharkhand Budget Analysis 2026–27, and budget summaries (see Sources). Figures to be confirmed against the Budget at a Glance.

The evidence of leakage

The Comptroller and Auditor General's revenue-sector audits of Jharkhand show the same pattern year after year: revenue is lost not for lack of rules but because records that should agree are never compared.

- **₹12,737 crore** of under-assessment, short levy or loss of revenue found in 45,954 test-checked cases across sales tax, excise, vehicles, land revenue, stamps, electricity duty and mining (CAG Report No. 4 of 2016); departments accepted ₹11,676 crore of it.
- **₹446 crore** short levy of royalty from undervaluing coal rejects in a colliery's returns, and ₹143.5 crore from district offices applying incorrect royalty rates (CAG revenue-sector reports).
- **₹12.4 crore** short levy of stamp duty and registration fees because mining leases were registered without checking the royalty on the approved mining plan.
- **₹70.9 crore** potential loss from 368 non-operational sand ghats, plus short-levied royalty and unrealised dead rent, in the CAG's 2025 report on minor minerals, which also found minor-mineral revenue falling from ₹1,082 crore (2017–18) to ₹698 crore (2021–22).
- **Excise:** since the 2025 policy, wholesale supply stays with JSBCL and retail sale is with private licensees, creating two records that can be reconciled continuously for every licensee.

WHAT THESE HAVE IN COMMON

Each finding is a disagreement between two records the state already holds: a royalty rate and the notified rate; a lease's royalty and the stamp duty on its deed; a lessee's returns and its despatches. Audit finds them years later, one sample at a time. A platform that compares the records continuously finds them while they can still be recovered.

2 WHAT JHARKHAND HAS TODAY: A COMPARATIVE STUDY

Jharkhand has invested well in departmental systems. The question for revenue leakage is whether any of them compares one department's records with another's for the same taxpayer and turns the disagreement into a verified rupee figure.

Department	Existing system	What it does well	What it cannot see
Commercial Taxes	GST system (GSTN), state tax portal; national BIFA / GST Prime analytics	Registration, returns, e-way bills; GST-only risk outputs from the national analytics	Electricity use, property, vehicles, stamps, mining or excise of the same taxpayer; state access to BIFA is limited to a few officers
Mines & Geology	JIMMS (upgraded May 2025)	Leases, e-challans and transit permits; shares data with ports, railways, Commercial Taxes, Income Tax and treasury	Whether the lease deed was stamped on the right royalty, or whether a lessee's GST and power use match reported extraction
Registration	NGDRS / Jharnibandhan	Online valuation, e-stamp, e-payment and appointments	Whether the parties' turnover, land use or repeat pattern suggests undervaluation
Land records	Jharbhoomi	Records of rights and online mutation	Commercial activity on land not converted for that use; delays between deed and mutation
Transport	VAHAN / SARATHI (NIC)	Registration, permits, tax payment	Commercial use of privately registered vehicles; fleet size against declared business scale
Excise	Excise portal; JSBCL wholesale	Licences, wholesale supply	Retail sales against duty-paid quantity and against the licensee's GST
Energy	JBVNL billing	Connections, consumption, billing	Dormant GST registrations drawing commercial power; tariff misclassification
Municipal	SUDA municipal services portal	Holding, water and trade tax payment	Unassessed floors or commercial use visible in other records

How other tools compare

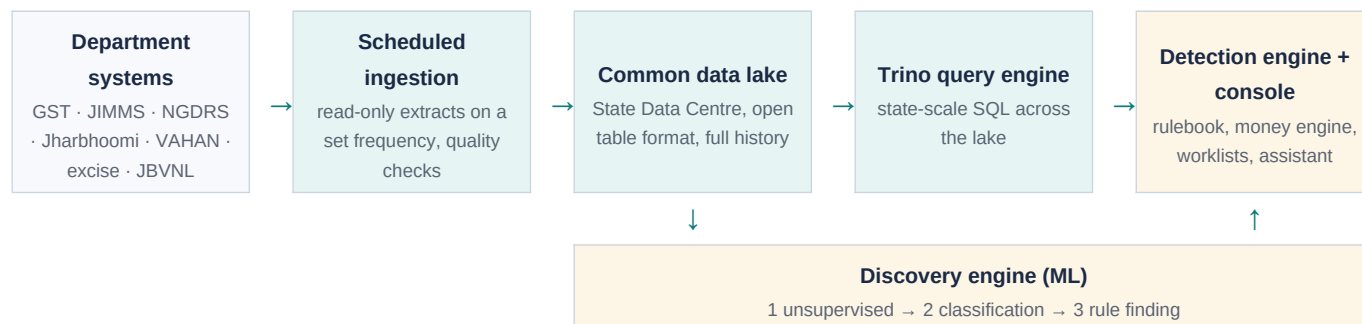
Capability	National GST analytics (BIFA)	JIMMS	Departmental MIS	TrueYield
Covers GST	Yes	No	Own dept	Yes
Covers mining, stamps, land, vehicles, excise, power	No	Mining only	Own dept	Yes
One view of a taxpayer across departments	No	No	No	Yes
Rupee figure with its arithmetic and statutory rate	Partly	Partly	No	Yes
Evidence separated from leads	No	No	No	Yes
Officer worklist and recovery tracking	Partly	No	Partly	Yes
Fully controlled by the state	No (GSTN)	Yes	Yes	Yes

CONCLUSION OF THE STUDY

Based on public information, Jharkhand has strong departmental systems but no platform that compares them. Other states are moving GST-first (Karnataka with IIT Hyderabad; Telangana's push for AI), but none, to our knowledge, compares GST with mining, stamps, land, vehicles, excise and power for the same taxpayer. TrueYield would sit on top of JIMMS, NGDRS, Jharbhoomi, VAHAN, JBVNL and the tax systems, reading from each and changing none, and would complement the national GST analytics. To be confirmed with the Finance and Commercial Taxes Departments in Phase 0.

3 THE PROPOSED SOLUTION

TrueYield has four layers. Data is extracted read-only from each department on a schedule and held in one place; a query engine makes it searchable at state scale; the detection engine applies the rulebook; and the discovery engine looks for patterns the rulebook does not yet cover.



- **Common data lake:** read-only extracts from each department on a set frequency (daily or weekly), quality-checked on arrival and kept with full history in the State Data Centre.
- **Trino query engine:** a distributed SQL engine that runs every check across the lake at state scale and can also query a department's database directly where an extract is not yet possible.
- **Detection engine:** the rulebook of checks, the money engine that computes each rupee once with its derivation and grade, and the entity resolution that links records for the same taxpayer (PAN, GSTIN, or name, address and mobile where no shared key exists).

Detection engine: check families adapted to Jharkhand's registers

Family	Jharkhand data	Examples of what is checked
GST & allied	GST returns, e-way bills	Turnover falling while power use rises; ITC far above sector norms; turnover split across GSTINs to stay under thresholds; circular trading
Mines & minerals	JIMMS leases, e-challans, royalty	Royalty due vs paid at the notified rate; extraction beyond permit; operations after lease expiry; dead rent unpaid; idle leases with power drawn
Registration & stamps	NGDRS / Jharnibandhan	Document type set to a lower-duty class; repeat parties undervaluing; lease deeds stamped below the royalty-based value
Land records	Jharbhoomi	Commercial activity on non-converted land; long gap between deed and mutation
Motor vehicles	VAHAN	Road tax unpaid; commercial use of privately registered vehicles; permit fee leakage
State excise	Excise portal, JSBCL	Sales vs duty-paid quantity; licence fee shortfall; excise sales vs GST declared
Power & utilities	JBVNL	Dormant GST registration with active commercial connection; commercial use billed as domestic
Cross-department	All of the above	Business growing in assets but not in tax; the same taxpayer non-compliant in several departments; mining extraction vs transport movement

What officers and leaders see

- **Taxpayer 360:** every register's view of one taxpayer, with findings first and each figure traceable to its source record.
- **Verification worklist:** ranked by district, circle and department, assigned to named officers, with status from assignment to recovery.
- **Recovery drill-down:** state > district > circle, separating recurring revenue from one-time recovery.
- **Leadership dashboard:** what has been found, verified and recovered, and how much rests on evidence rather than leads.
- **Ask-a-question assistant:** officers ask about a taxpayer in plain Hindi or English and get answers that cite the records, from an AI model hosted inside the State Data Centre.
- **Policy simulator:** estimate the effect of a rate or threshold change before it is notified.

4 THE DISCOVERY ENGINE: FINDING THE RULES NO ONE HAS WRITTEN YET

The detection engine finds what its rules describe. Leakage also takes forms no rule yet describes. The discovery engine uses three layers of machine learning over the common data lake to find those forms and turn them into rules that the detection engine can then run every day.

Layer	What it does	What it produces
1. Unsupervised discovery	Looks across all departments' records for taxpayers and behaviour that stand apart from their peers (anomaly detection and clustering), with no prior labels.	Groups of unusual taxpayers and the features that set them apart
2. Classification	Learns from officers' verified outcomes (confirmed leakage or a legitimate explanation) which unusual patterns are real, and scores new cases accordingly.	Patterns ranked by how often they turn out to be genuine
3. Rule finding	Turns each confirmed pattern into a short, explainable rule: plain conditions on the taxpayer's records with clear thresholds.	Candidate rules in the same form as the rulebook

FROM A PATTERN TO A RULE

- Each candidate rule is back-tested on historical records and reviewed by departmental officers and our domain team.
- Only approved rules enter the detection engine, each with a statutory basis, a grade and an evidence or lead ceiling like every other rule.
- The ML never raises a case against a taxpayer directly; every case an officer acts on comes from a rule that can be explained in terms of the taxpayer's own records.
- The engine learns from verified outcomes, so it starts after the pilot has produced them and improves with every monthly cycle.

SAFEGUARDS BUILT IN

- A discrepancy is not an evasion. The output is a verification list; nothing is assessed, billed or demanded by the platform.
- Evidence (both records are the state's own, keyed to the same taxpayer) is kept apart from leads (patterns and benchmarks), and a lead is never promoted to evidence automatically.
- Every statutory rate sits in one table that the department signs off; no rupee is computed from an unsigned rate.
- Read-only access, role-based permissions, audit logs, hosting in the State Data Centre, DPDP Act 2023 compliant.

5 WHAT THE CAG FOUND, AND HOW TRUEYIELD WOULD CATCH IT

CAG finding in Jharkhand	The TrueYield check that compares the two records
Royalty short-levied by undervaluing coal rejects in returns (₹446 crore)	Extraction vs royalty gap: value and quantity in the lessee's returns compared with JIMMS despatches and the notified basis
Incorrect royalty rates applied by district offices (₹143.5 crore)	Royalty recomputed from the signed-off rate table for each mineral and lease; any difference is flagged with both rates shown
Mining leases registered without verifying royalty; short stamp duty (₹12.4 crore)	Cross-department: the lease's annual royalty in JIMMS compared with the value on which its deed was stamped in NGDRS
Short levy of royalty and unrealised dead rent (2025 report)	Repeated short payments; dead rent due vs paid per lease and year
Non-operational sand ghats and idle leases (₹70.9 crore potential loss)	Lease capacity underutilised; inactive lease with power or transport activity nearby

These are illustrations of fit, not claims about current cases. Whether the same patterns exist today is what the pilot will measure.

6 APPROACH AND METHODOLOGY

The detection engine already exists. The work is to license and adapt it to Jharkhand's statutes, build the state-scale data platform, prove its value on real records, and then add the discovery engine once verified outcomes are available to learn from.

Phase	Duration	What we do	Output
0. Mobilise	4 weeks	Data-sharing order from the Finance Department; read-only access to Commercial Taxes, Mines & Geology and Registration; confirm existing tools; department sign-off on the rate table.	Signed data compact and rate table
1. Pilot	3 months	Set up the data lake and Trino in the State Data Centre; scheduled ingestion from the three departments; adapt checks to Jharkhand law; run in three districts (proposed: Ranchi, Dhanbad, East Singhbhum); officers verify the top cases.	Verified findings, measured hit rate, labelled outcomes
Decision gate	2 weeks	Pilot results reviewed with the Finance Secretary; proceed only if verified findings justify it.	Go / no-go
2. Rollout	6 months	Connect Transport, Excise, JBVNL and land records; all 24 districts; officer training; security audit; leadership dashboard.	Statewide platform
2a. Discovery	Months 2–6 of rollout	Build the three ML layers on pilot outcomes; first candidate rules reviewed and added to the rulebook.	Discovery engine; first new rules
3. Run	Ongoing	Scheduled ingestion; rule and rate updates; monthly discovery cycle and model retraining; monthly findings and recovery report.	Monthly recovery report and new rules

Optional: a municipal edition (holding, water and trade tax) already exists from our earlier Jharkhand work and can be added for Ranchi and two other urban local bodies.

7 BENEFITS

For the Government

- Revenue the state is already owed, found while it can still be recovered rather than years later in audit
- One view of every taxpayer across departments
- Officers' time directed to the highest-value, best-evidenced cases
- Audit readiness: every figure traceable to its record and rate
- Evidence for policy: what a rate or threshold change would yield
- A rulebook that keeps growing as the discovery engine finds new patterns

For honest taxpayers and businesses

- Fewer blanket inspections; scrutiny targeted at real discrepancies
- A level playing field against under-reporting competitors
- Clear, record-based reasons whenever a case is raised

SCALE, NOT A FORECAST

If verified recoveries reached just 0.1% of own tax revenue (about ₹46 crore a year), that would be more than twenty times the platform's full one-time cost of ₹1.97 cr. How much leakage exists in Jharkhand today is not known by anyone; measuring it on real records is the purpose of the pilot.

8 COST

One-time component	Pilot	Rollout	Total
TrueYield platform licence: detection engine, 85-check library, console, assistant, policy simulator (existing product; perpetual, state-wide)	₹12.5 lakh	₹37.5 lakh	₹50 lakh
Data platform: common data lake, Trino query engine, scheduled ingestion, data-quality checks, department connectors	₹21.5 lakh	₹25 lakh	₹46.5 lakh
Adaptation and deployment: Jharkhand rules and rates, entity resolution, Hindi, training, security audit	₹22.5 lakh	₹35 lakh	₹57.5 lakh
Discovery engine: three ML layers (unsupervised, classification, rule finding) with review workflow	—	₹42.5 lakh	₹42.5 lakh
One-time total	₹56.5 lakh	₹1.4 cr	₹1.97 cr

Recurring and optional	Amount
Infrastructure: servers, data lake storage, GPU server for the AI models, network, backup (see Annex)	<i>Provided by the State Data Centre</i>
Annual maintenance: support, upgrades, operations in the State Data Centre, rule updates, monthly discovery cycle, helpdesk	₹47.5 lakh / yr
Optional: municipal edition for 3 urban local bodies (one-time)	₹27.5 lakh

Cost assumptions

- The detection engine and console are an existing product, licensed perpetually for use across Government of Jharkhand departments; the pilot licence fee is credited against the full licence. The core product's source code is placed in escrow.
- The state owns its data, its Jharkhand-specific rules and rates, the department connectors and the rules the discovery engine finds for it.
- Hosted entirely in the Jharkhand State Data Centre (JAP-IT), which provides all infrastructure: servers, data lake storage, the GPU server, network, security, backup and disaster recovery (see Annex). Kirnova does not supply or charge for infrastructure. All data and AI models (open-source, on the state's servers) stay inside the data centre.
- Prices cover delivery, integration, project management, on-site support in Ranchi, training, and contingency for data-access and data-quality work. All amounts exclude GST and are valid for 90 days from the date of this proposal; changes in scope are priced separately.

9 KEY RISKS AND SAFEGUARDS

Risk	How we handle it
Departments reluctant to share data	Read-only access under a Finance Department order; data stays in each department; departments see their own findings first.
Limited access to GST data held by GSTN	Start with the returns and e-way bill data available to the Commercial Taxes Department; national analytics outputs are used alongside, not replaced.
Records do not share a common key	PAN and GSTIN where available; name, address and mobile matching elsewhere, with every inferred link graded and shown as a lead.
Too many false alarms	Evidence separated from leads; ranked lists; officers verify before any action; thresholds tuned on pilot results.
Rates or rules change	One signed-off rate table; updates coded within two weeks of notification.
Data extracts load department systems	Scheduled off-peak extracts, read replicas or change capture where available; frequency agreed with each department.
ML proposes an unsound rule	Back-testing and officer review before any rule is used; the ML never raises a case directly.
Privacy and misuse	Role-based access, audit logs, State Data Centre hosting, DPDP Act 2023 compliance.

10 HOW WE WILL WORK WITH THE GOVERNMENT

Start small, prove value	A 4-month pilot on real records in three districts; statewide rollout only after a decision gate.
Transparent procurement	Engagement through GeM, JAP-IT or an empanelled system integrator.
Data stays with the state	Everything, including the data lake and the AI models, runs in the Jharkhand State Data Centre; no data goes to any external cloud or AI service.
No vendor lock-in	Open-source data components (Trino, open table formats); product source code in escrow; the state owns its data, rules and connectors; full documentation and training.
Pay for outcomes	Milestones tied to measurable results: verified findings, cases assigned, recoveries tracked.

Sources (public information, accessed September 2026; status descriptions to be confirmed with the Departments)

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Annex HOSTING REQUIREMENTS (PROVIDED BY THE STATE DATA CENTRE)

TrueYield runs entirely inside the Jharkhand State Data Centre. The following infrastructure is provided by the State Data Centre (JAP-IT); Kirnova installs, configures and operates the software on it. Sizing is for the statewide rollout with seven departments and is confirmed with JAP-IT in Phase 0; the pilot needs roughly half.

Component	Requirement
Data lake storage	S3-compatible object storage on the data centre's storage, 20 TB usable to start and expandable, holding department extracts with full history
Query engine (Trino)	1 coordinator: 16 vCPU / 64 GB RAM; 4 workers: 16 vCPU / 128 GB RAM each
Ingestion and scheduling	2 virtual machines, 8 vCPU / 32 GB RAM (scheduler and department connectors)
Application and console	2 virtual machines, 8 vCPU / 32 GB RAM, load-balanced
Database	2 virtual machines, 8 vCPU / 32 GB RAM, 1 TB SSD each (PostgreSQL, primary and standby)
AI and machine-learning server	1 GPU server with 2× NVIDIA L40S (48 GB) or A100 (80 GB) or equivalent, 32 vCPU / 256 GB RAM, for training the discovery engine and running the in-house assistant model
Staging environment	4 virtual machines, 8 vCPU / 32 GB RAM, and 2 TB storage
Network	Secure read-only connectivity from the data centre to departmental systems over the state network; firewall rules; SSL certificates
Backup and disaster recovery	As per the State Data Centre's standard policy